Transformers Learn Generalizable CoT Reasoning via Gradient Descent

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Yale Statistics & Data Science

MIT IDSS Seminar
2025

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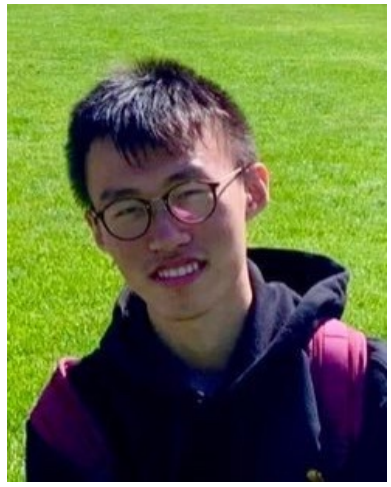

Zixin Wen

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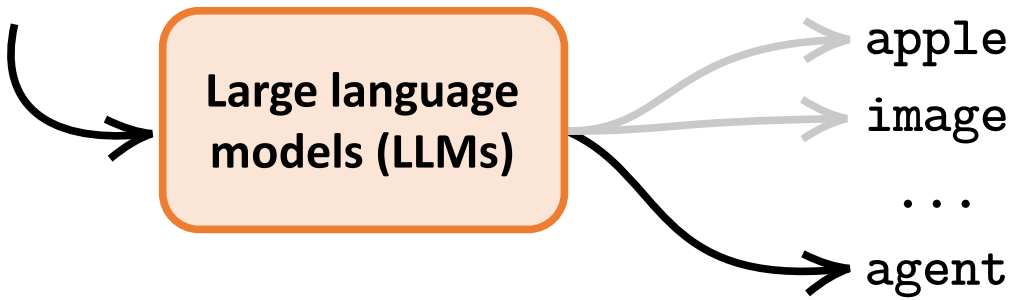
Large language models

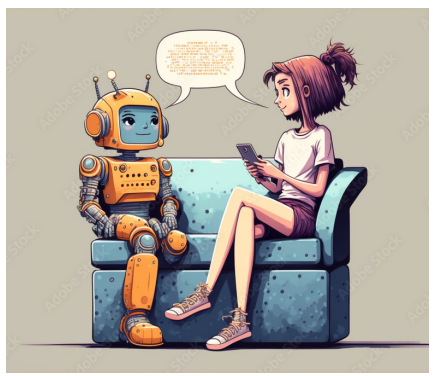
The language models predict the next word based on previous words.

Prompt: Explain reinforcement learning (RL).

Answer: Reinforcement learning (RL) is a type of machine learning where an...



Applications in translation, Q&A, summarization...



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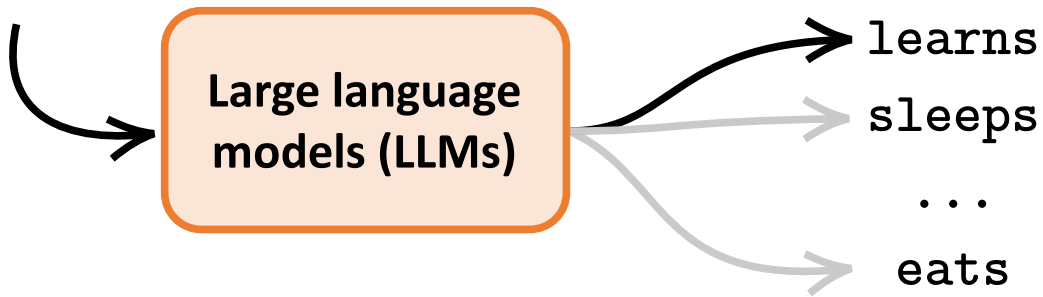

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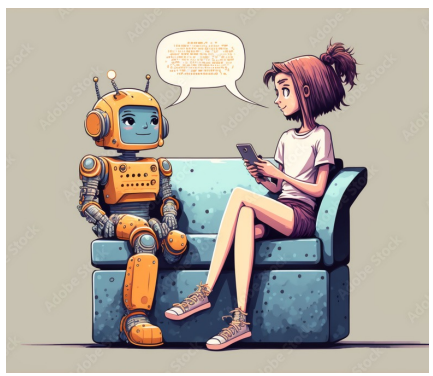
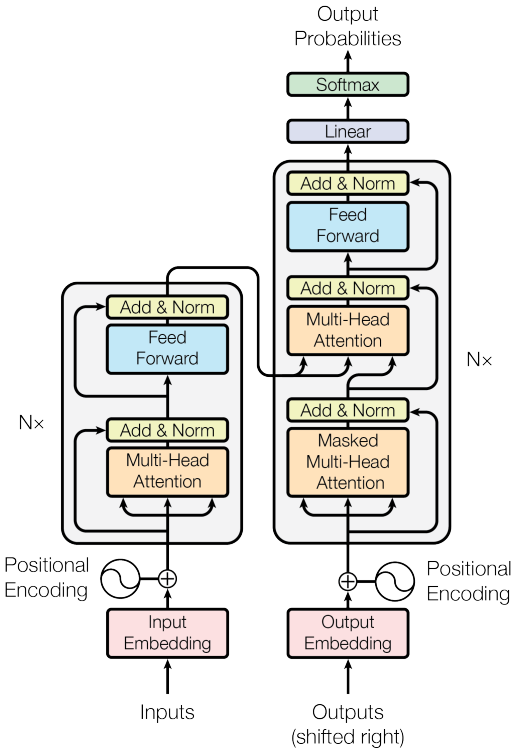


Transformers as the backbone architecture



Attention is all you need

A Vaswani, N Shazeer, N Parmar… - Advances in neural …, 2017 - proceedings.neurips.cc

The dominant sequence transduction models are based on complex recurrent orconvolutional neural networks in an encoder and decoder configuration. The best …

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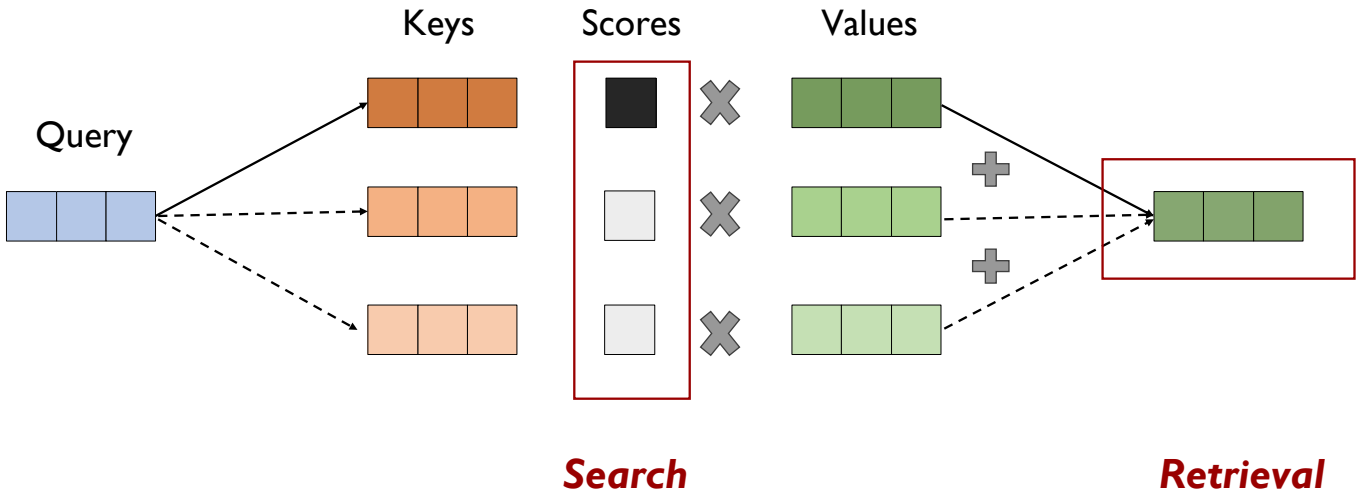
Attention head

Attention head: For each query q, it matches against all the key-value pairs (k_i, v_i) and compute the value v:

$$\text{head}(q) = \sum \alpha_i v_i, \quad \text{where} \quad \alpha_i = \frac{\exp(q^\top k_i/\sqrt{d})}{\sum_u \exp(q^\top k_u/\sqrt{d})}.$$



Attention searches and retrieves the values of relevant tokens

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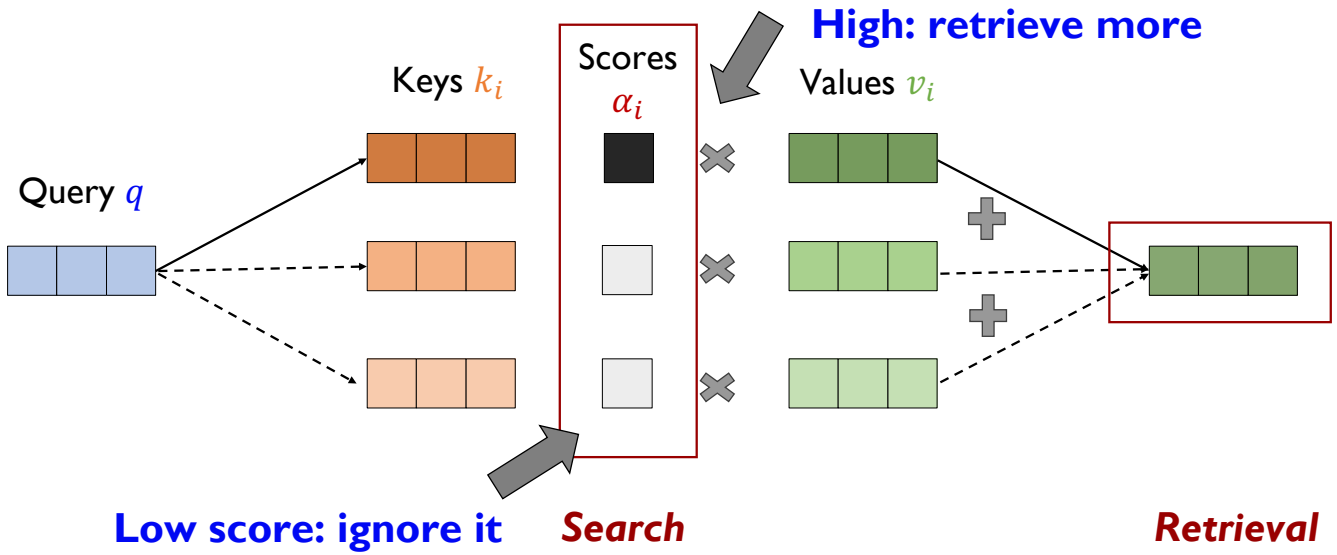
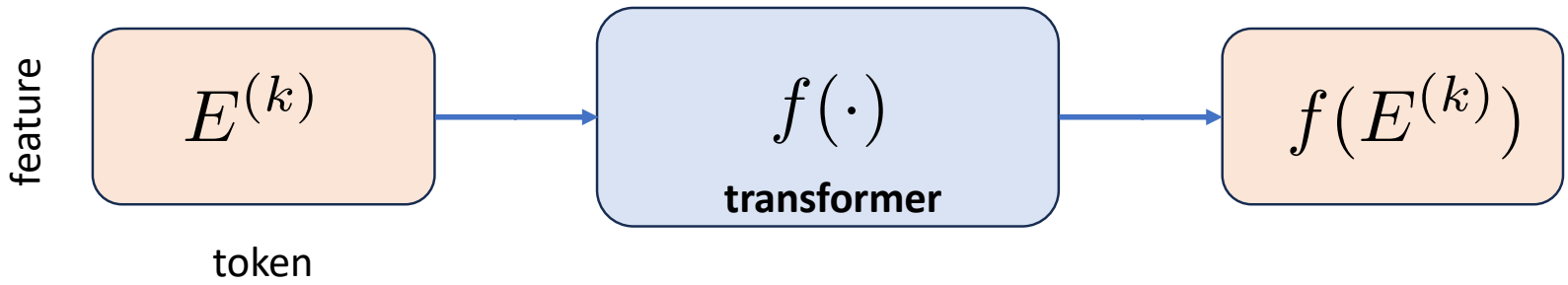


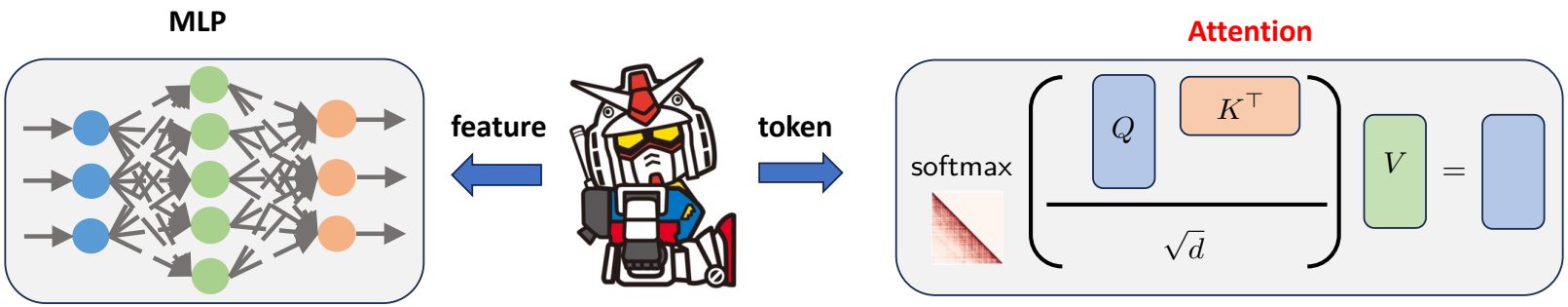
Attention searches and retrieves the values of relevant tokens

Transformers as the backbone architecture

Sequence-to-sequence:



Transformer block = attention + MLP



Transformers as the backbone architecture

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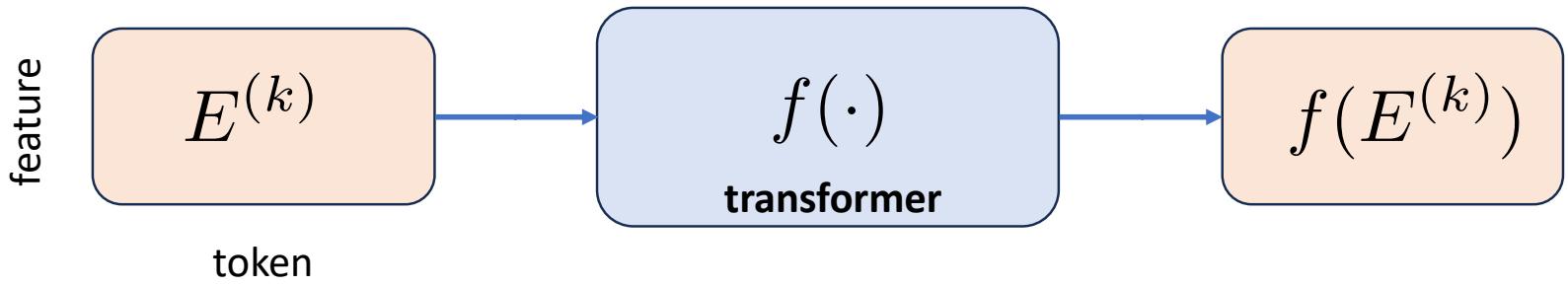
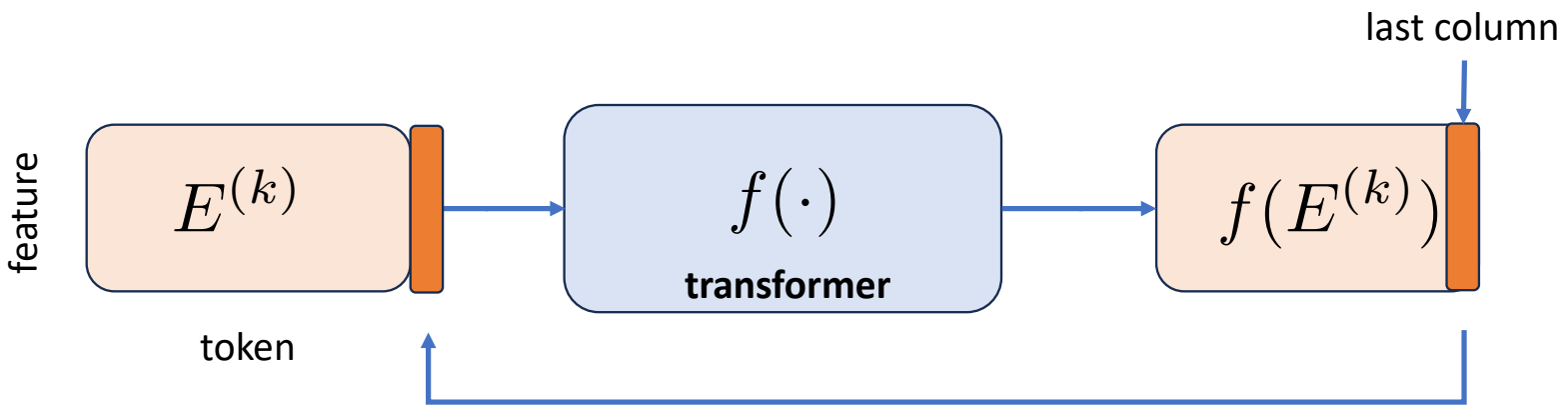
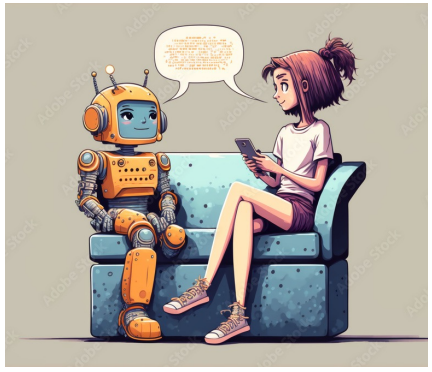
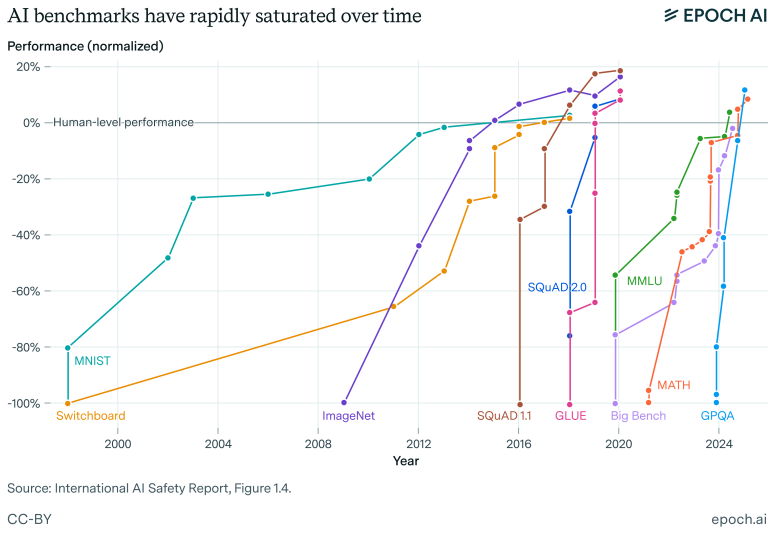


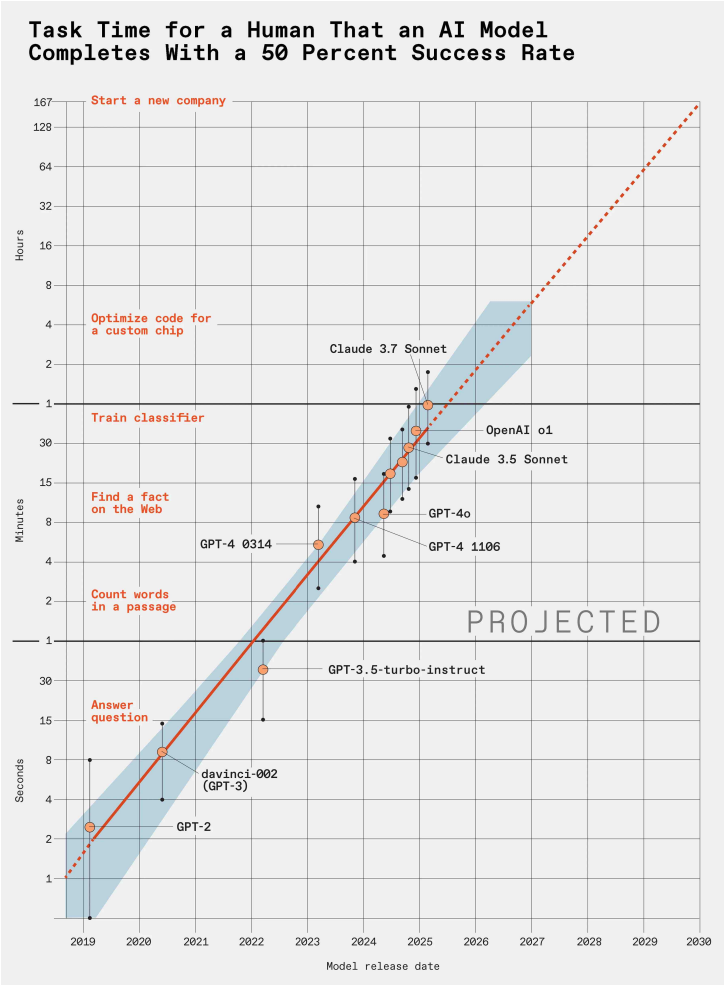
Autoregressive decoding:



The rise of large language models







LLM benchmarking shows capabilities doubling every **7 months**.

`https://spectrum.ieee.org/large-language-model-performance`

Chain-of-thought (CoT)

LLMs solve harder questions by reasoning **step-by-step**: generate intermediate reasoning steps before inferring an answer.*

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

<Question→Answer>

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

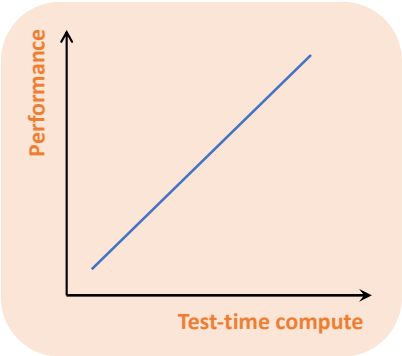
Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

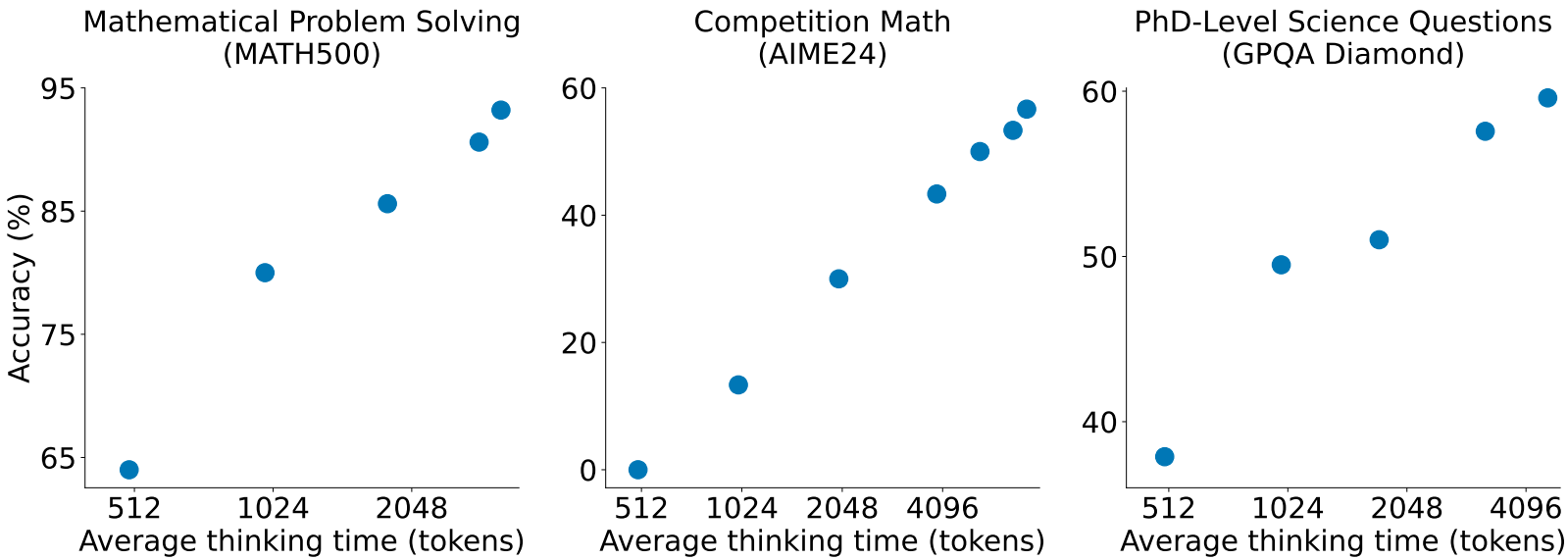
<Question→Rationale→Answer>

*Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.

Test-time scaling

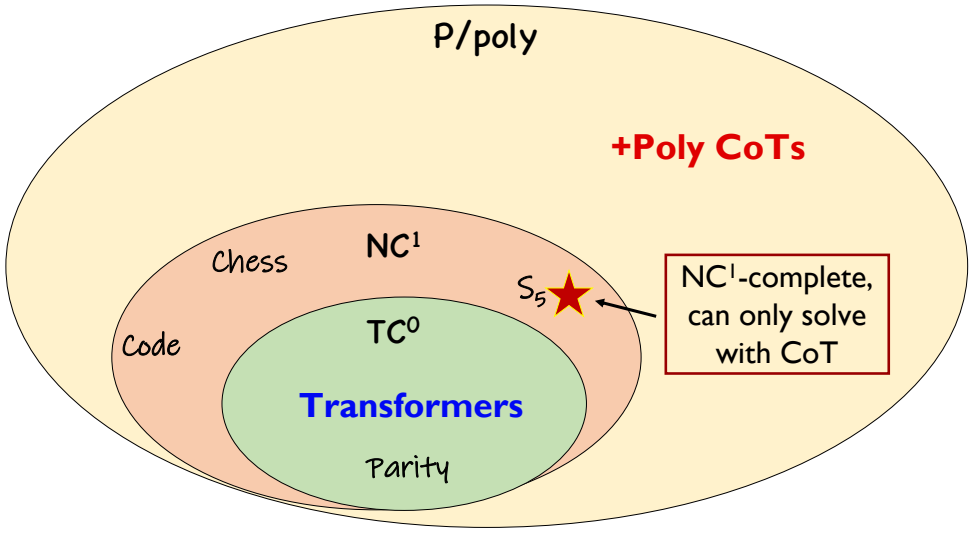


Reasoning performance improves with more test-time compute (e.g., thinking longer).



(Figure credit: s1: Simple test-time scaling)

Expressive power of CoT



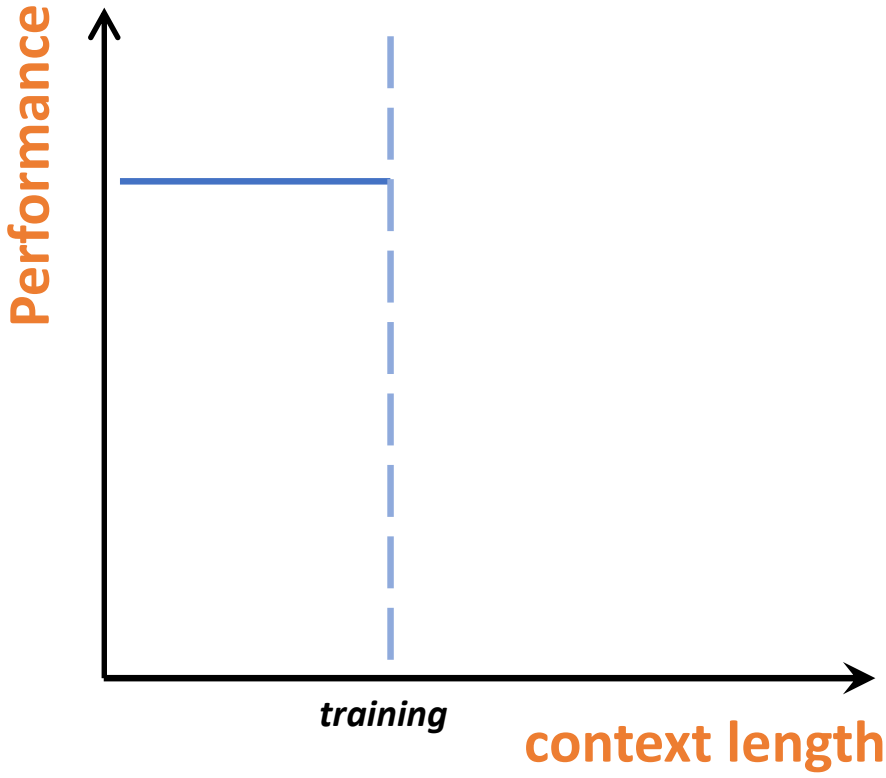
assuming $TC^0 \neq NC^1$

Expressiveness: (Li et al., 2024; Merrill et al., 2023; Feng et al., 2023...)

- without CoT $\subset \mathsf{TC}^0$;
- with linear CoTs $\supseteq \mathsf{NC}^1$;
- with polynomial CoTs $\supseteq P/\text{poly}$.

With increasing CoT lengths, transformers become more powerful!

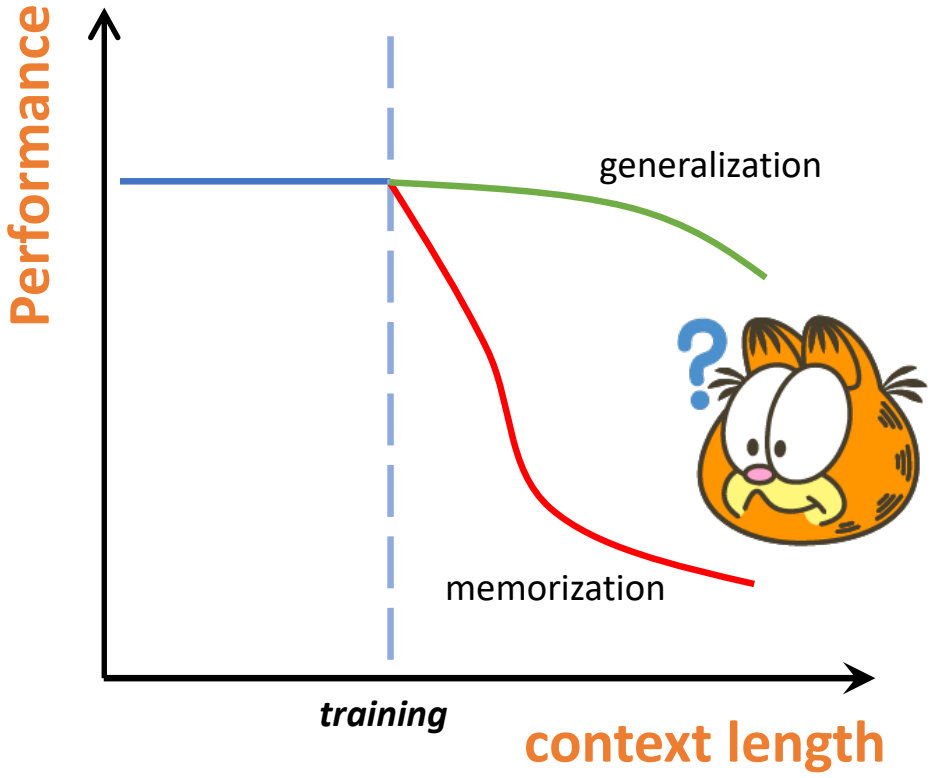
Length generalization from training perspective



Can transformers trained on shorter tasks be generalized to longer ones?

No optimization guarantees *beyond* TC^0.

Length generalization from training perspective



Can transformers trained on shorter tasks be generalized to longer ones?

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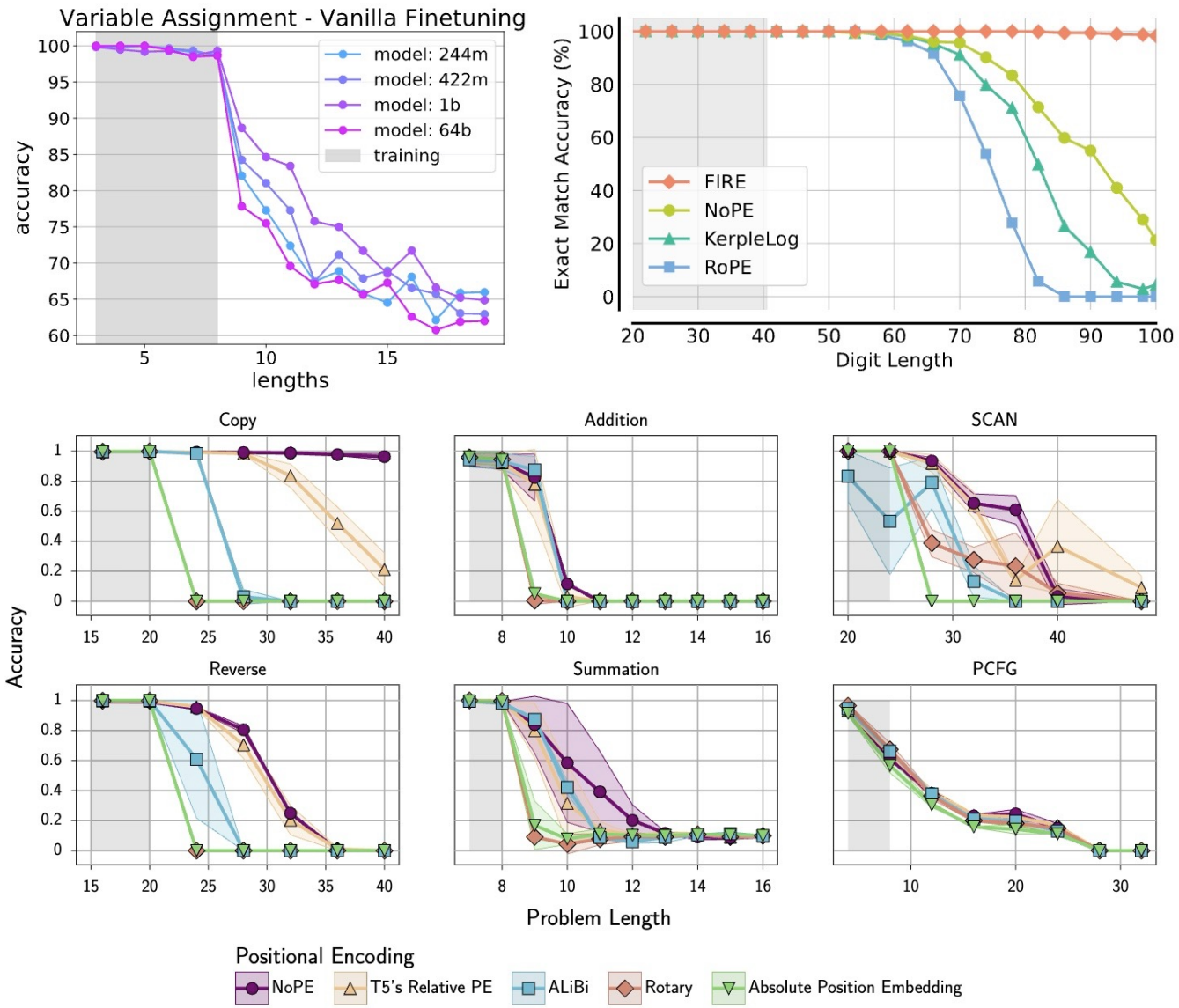
Empirical evidence is mixed



(Figure credit: Anil et al., 2022; Kazemnejad et al., 2023; Zhou et al., 2024)

This talk: training dynamics of CoT reasoning

Why do *trained* transformers perform multi-step reasoning?

Optimization

Generalization

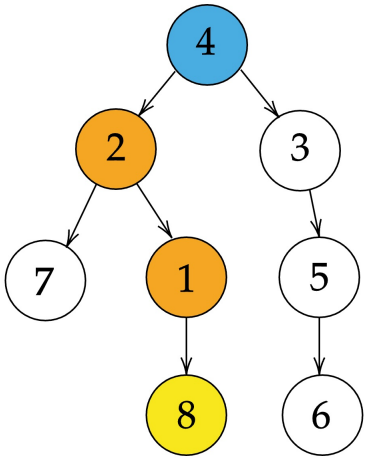
- What does gradient descent converge to?
- What is the trained attention at convergence?

- Will trained transformers generalize to unseen tasks?
- Will trained transformers generalize to longer lengths?

Take-away: one-layer transformers with CoT training can solve reasoning tasks that are otherwise out of reach

This talk: two case studies with one-layer transformers

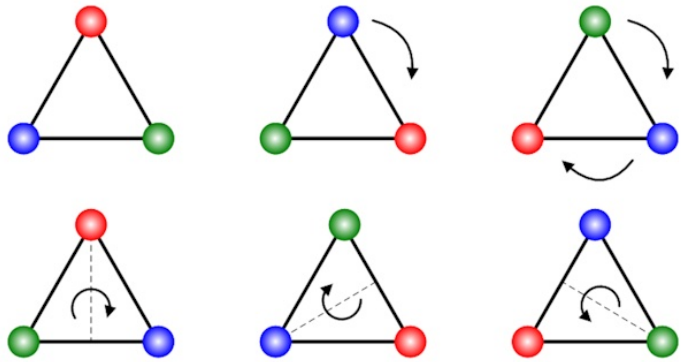
symbolic reasoning with multi-head attention



Path-finding in a tree

(Brinkman et al., 2024)

reasoning with group action with attention + MLP



LEGO

(Zhang et al., 2023)

Multi-head transformers learn symbolic multi-step reversal reasoning

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Yu Huang
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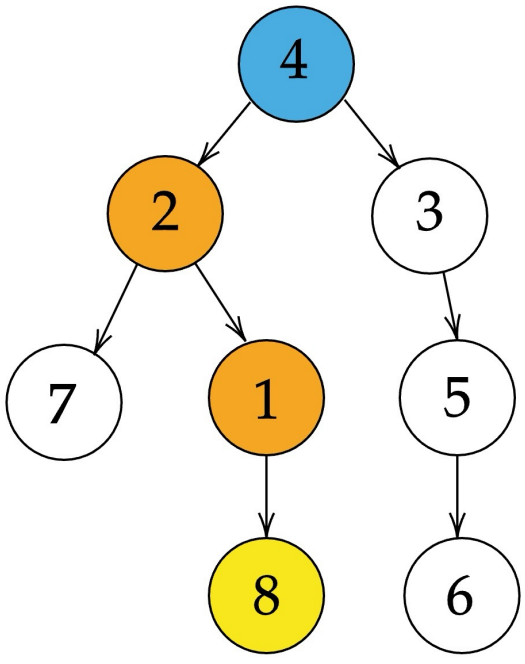
Yingbin Liang
OSU

"Multi-head Transformers Provably Learn Symbolic Multi-step Reasoning via Gradient Descent,"
T. Yang, Y. Huang, Y. Liang, Y. Chi, *NeurIPS 2025*

Pathfinding in trees



Prompt:

Edge set: $1 \to 8,\ 4 \to 2,\ 2 \to 7, 3 \to 5,$
$2 \to 1,\ 4 \to 3,\ 5 \to 6$

Goal: 8

Root: 4

Path-finding task:

Goal-to-root (backward):

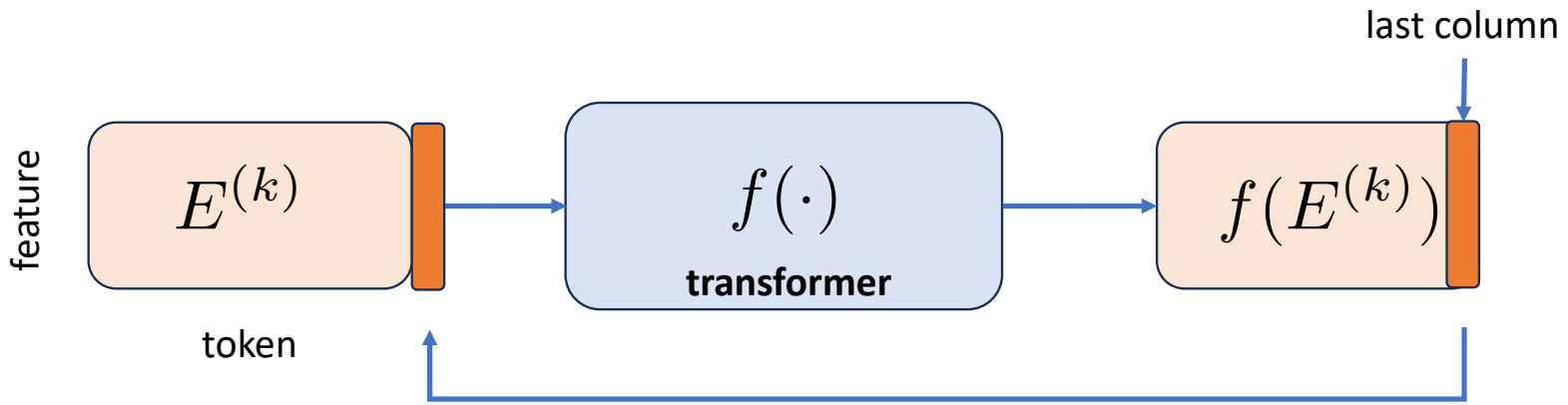
$8 \to 1 \to 2 \to 4$

Root-to-goal (forward):

$4 \to 2 \to 1 \to 8$

- **Backward** reasoning: find goal-to-root path; easy by backtracking.
- **Forward** reasoning: find the root-to-goal path; harder and needs to first solve backward task and then reverse it.

Autoregressive, multi-step, decoding



Embedding: each edge $e = (p(i), i) \in \mathcal{E}(\mathcal{T})$ is embedded by

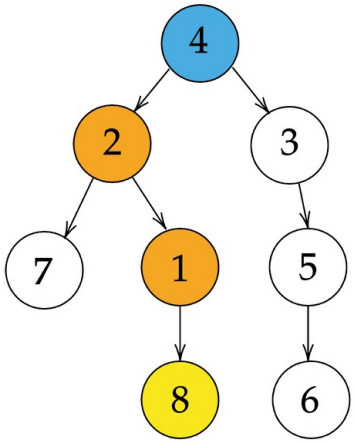
$$\begin{bmatrix} p(i) \\ i \end{bmatrix} \longrightarrow \begin{bmatrix} a_{p(i)} \\ a_i \end{bmatrix},$$

where a_i is the embedding of node i, $p(i)$ is the parent of node i.

Embedding of the input tree: (x_i, y_i)'s are the edges

$$E(\mathcal{T}) = \begin{pmatrix} X(\mathcal{T}) \\ Y(\mathcal{T}) \end{pmatrix} = \begin{pmatrix} x_1 & \dots & x_{l(\mathcal{T})} & a_0 & a_{g(\mathcal{T})} \\ y_1 & \dots & y_{l(\mathcal{T})} & a_{r(\mathcal{T})} & a_0 \end{pmatrix} \in \mathbb{R}^{2d_1 \times (l(\mathcal{T})+2)}$$

Single-head attention for one-step backward reasoning



$$E^{(0)} = \begin{bmatrix} X^{(0)} \\ Y^{(0)} \end{bmatrix} = \left[\begin{array}{ccccccc|c|c} 1 & 4 & 2 & 3 & 2 & 4 & 5 & 0 & 8 \\ 8 & 2 & 7 & 5 & 1 & 3 & 6 & 4 & 0 \end{array}\right] \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

edges r g

query

Attention searches the edge and retrieves the parent node

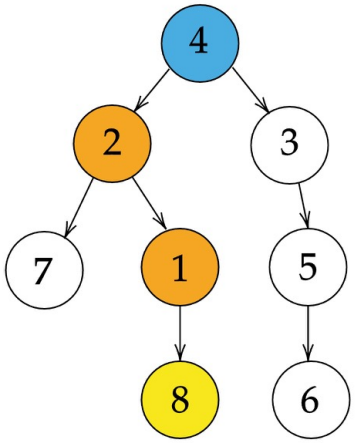
Attention output: simplified model with trainable B

$$\widehat{o}^{(k)} = \begin{pmatrix} X^{(k-1)} \\ Y^{(k-1)} \end{pmatrix} \cdot \text{softmax}\left(Y^{(k-1)\top} B x_{-1}^{(k-1)} \right),$$

where $x_{-1}^{(k-1)}$ is the last entry of $X^{(k-1)}$.

- Attend to $(p(n^{(k-1)}), n^{(k-1)})$, and output $(p(n^{(k-1)}), n^{(k-1)})$.

Single-head attention for one-step backward reasoning



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edges r g

search

query

Attention searches the edge and retrieves the parent node

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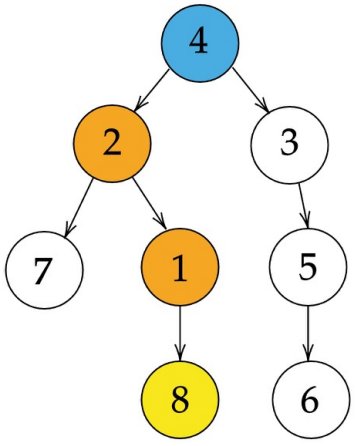

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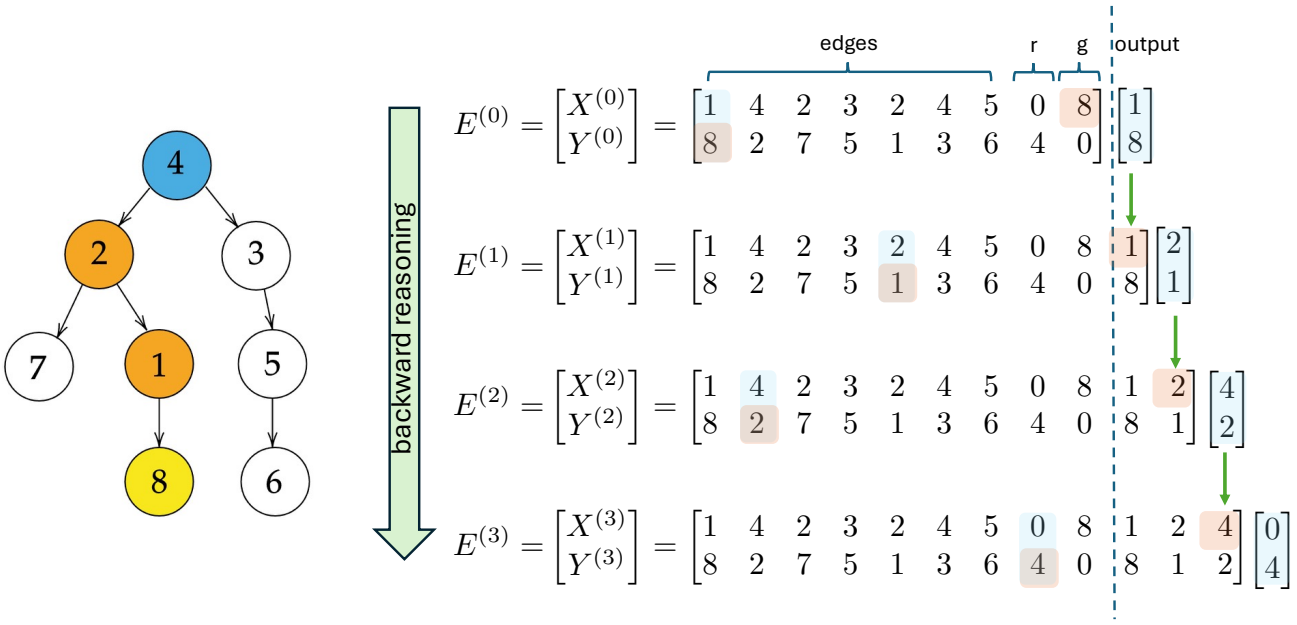
Single-head attention for backward reasoning

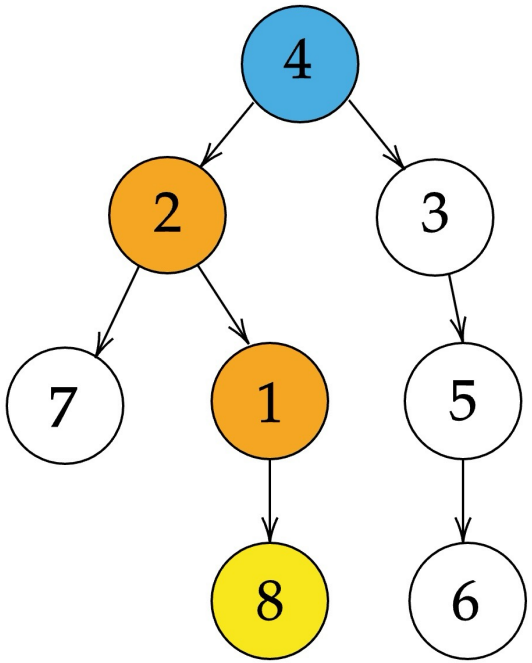


- Let $A := (a_1, \ldots, a_S) \in \mathbb{R}^{d_1 \times S}$ be the embedding matrix with linearly independent columns, and $a_0 = 0_{d_1}$.

Theorem (Construction for backward reasoning)

For any $\alpha \in \mathbb{R}$, there exists $B = B_\alpha \in \mathbb{R}^{d_1 \times d_1}$ such that $A^\top B A = \alpha I_S$. Then for any tree $\mathcal{T}$, we have $\widehat{O}(\mathcal{T}; B) \to O(\mathcal{T})$ as $\alpha \to +\infty$.

Task switching for forward reasoning



Root-to-goal (forward):

$$8 \to 1 \to 2 \to 4 \to 2 \to 1 \to 8$$

Stage 1
backtracking

Stage 2
CoT Retrieval

Task switch: To implement forward reasoning, we need to solve and automatically switch between two subtasks:

- Backtracking: find the goal-to-root path and store it in CoT;
- CoT retrieval: retrieve the root-to-goal path from CoT.

Two-head attention for forward reasoning

Embedding of the input tree with two heads:

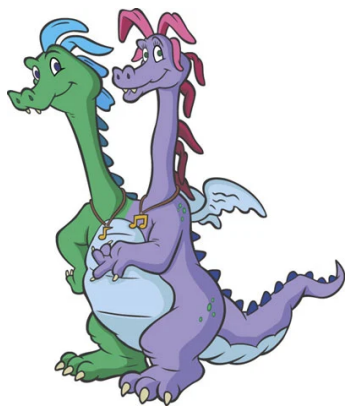
$$E_{\mathsf{r2g}}(\mathcal{T}) = \begin{pmatrix} X(\mathcal{T}) \\ Y(\mathcal{T}) \\ Z(\mathcal{T}) \end{pmatrix} = \begin{pmatrix} x_1 & \dots & x_{l(\mathcal{T})} & a_0 & a_0 \\ y_1 & \dots & y_{l(\mathcal{T})} & a_{r(\mathcal{T})} & a_{g(\mathcal{T})} \\ s_f & \dots & s_f & s_f & s_b \end{pmatrix},$$

where $s_f, s_b \in \mathbb{R}^{d_2}$ are stage token embeddings.

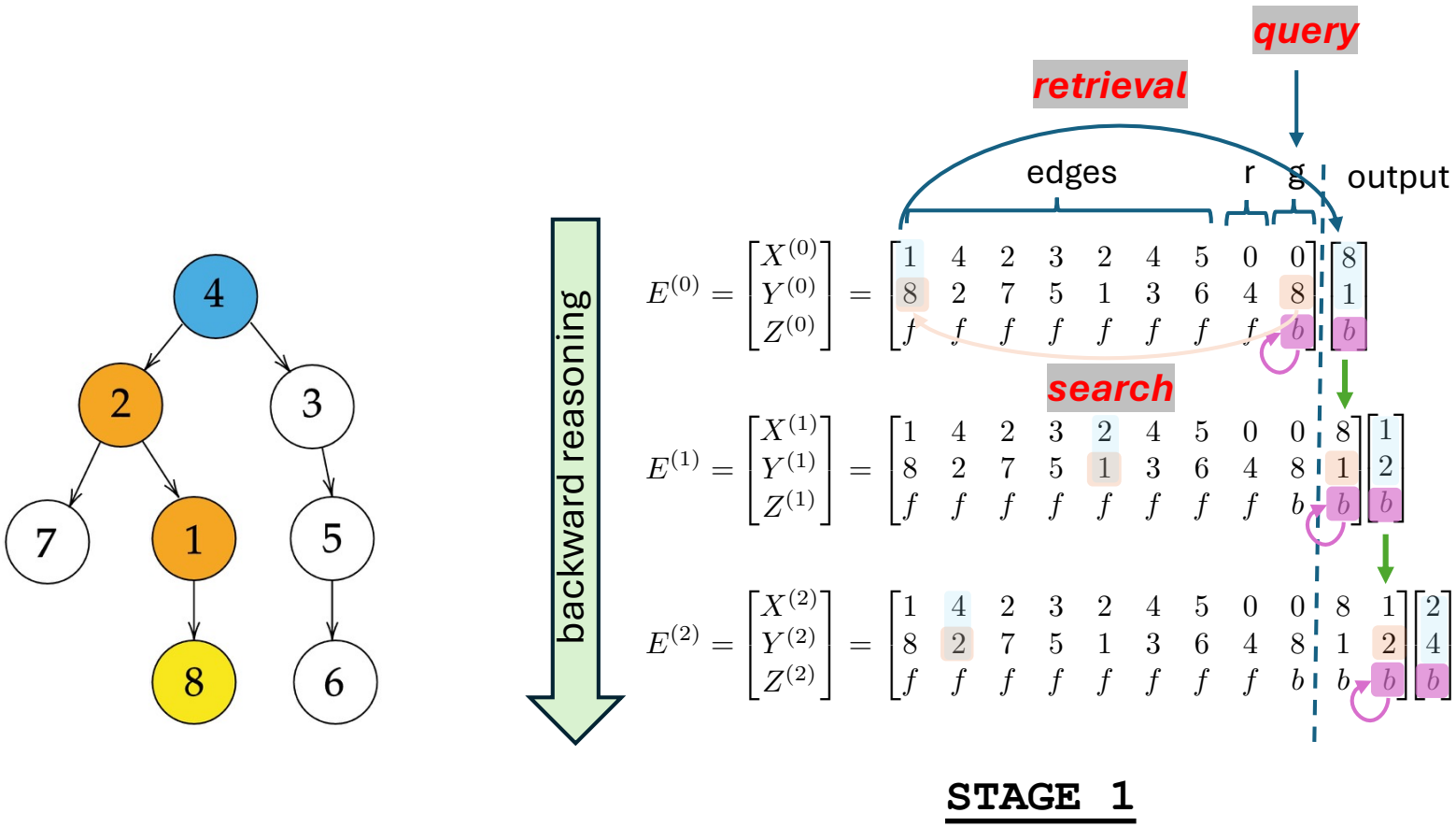
Attention output: simplified model with trainable $\{B_i, C_i\}_{i\in[3]}$:

$$\widehat{o}^{(k)} = \begin{pmatrix} \begin{pmatrix} Y^{(k-1)} \\ X^{(k-1)} \end{pmatrix} \cdot \mathsf{softmax}\left(X^{(k-1)\top} B_1 y_{-1}^{(k-1)} + Y^{(k-1)\top} B_2 y_{-1}^{(k-1)} + Z^{(k-1)\top} B_3 z_{-1}^{(k-1)} \right) \\ Z^{(k-1)} \cdot \mathsf{softmax}\left(X^{(k-1)\top} C_1 y_{-1}^{(k-1)} + Y^{(k-1)\top} C_2 y_{-1}^{(k-1)} + Z^{(k-1)\top} C_3 z_{-1}^{(k-1)} \right) \end{pmatrix}.$$

- The first head performs edge retrieval;
- The second head controls the task switch.

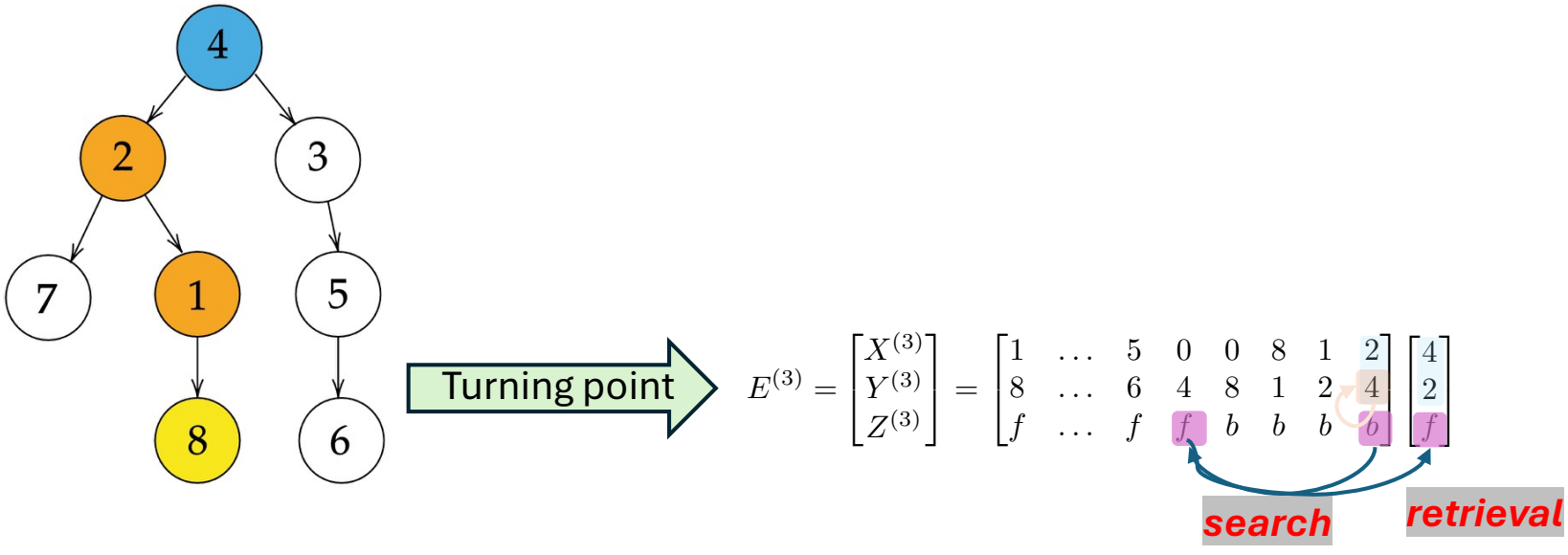


Two-head attention for forward reasoning: Stage 1



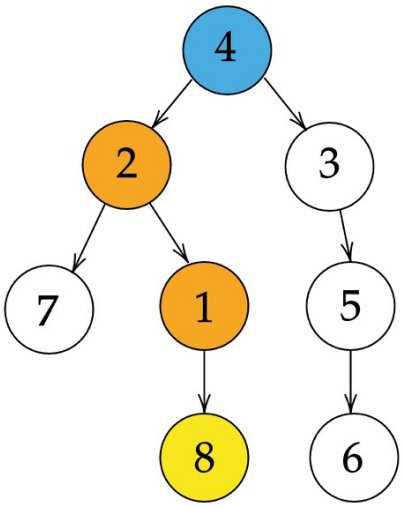
- First head: attend to $(p(n^{(k-1)}), n^{(k-1)})$, and output $(n^{(k-1)}, p(n^{(k-1)}))$.
- Second head: attend to and output (its own) stage signal s_b.

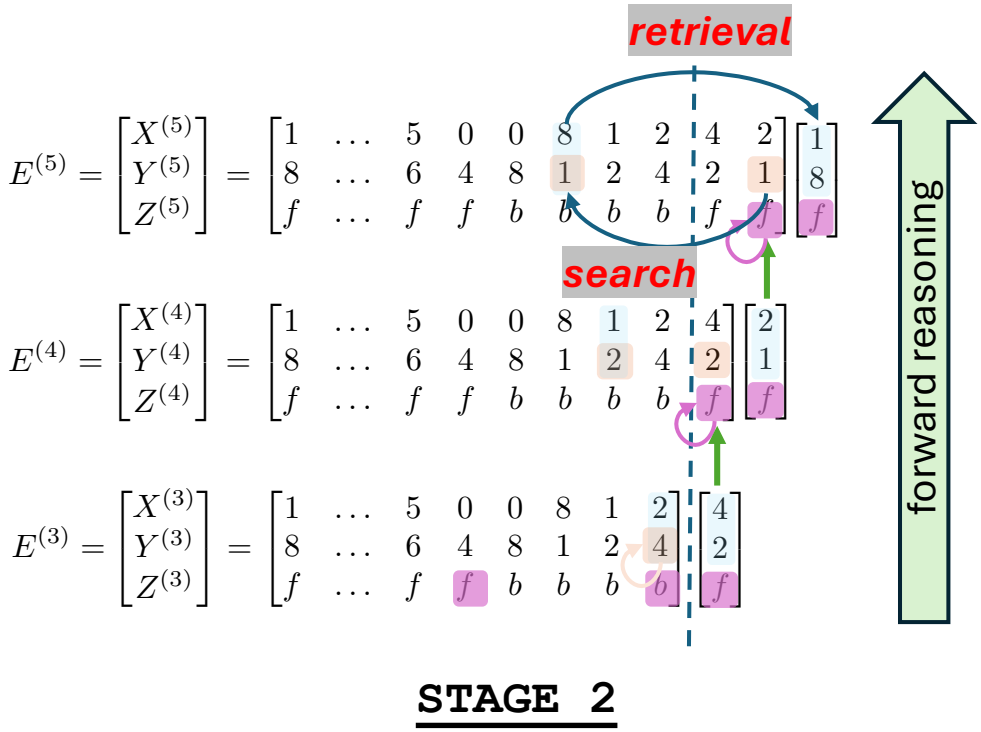
Two-head attention for forward reasoning: Turning Point



- First head: attend to $(y_{-1}^{(k-1)}, x_{-1}^{(k-1)})$ (itself) via null node embedding at root, and output $(x_{-1}^{(k-1)}, y_{-1}^{(k-1)})$.
- Second head: attend to and output stage signal s_f via null node embedding at root.

Two-head attention for forward reasoning: Stage 2





STAGE 2

- First head: attend to $(n^{(k-1)}, p(n^{(k-1)}))$ generated in *STAGE 1*, and output $(p(n^{(k-1)}), n^{(k-1)})$.
- Second head: attend to and output (its own) stage signal s_f.

Construction for forward reasoning

- Denote $\widetilde{A} = (a_0, a_1, \ldots, a_S) \in \mathbb{R}^{d_1 \times (S+1)}$ and $\widetilde{S} = (s_f, s_b)$ be the embedding matrix with linearly independent columns.

Theorem (Construction for forward reasoning)

There exist $\theta = \{B_i, C_i\}_{i \in [3]}$ *that satisfies*

$$\widetilde{A}^\top B_1 A = -a\alpha_1 \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \; A^\top B_2 A = \alpha_1 I_S, \; \widetilde{S}^\top B_3 \widetilde{S} = \alpha_1 \begin{pmatrix} -b_1 & b_2 \\ b_1 & -b_2 \end{pmatrix},$$

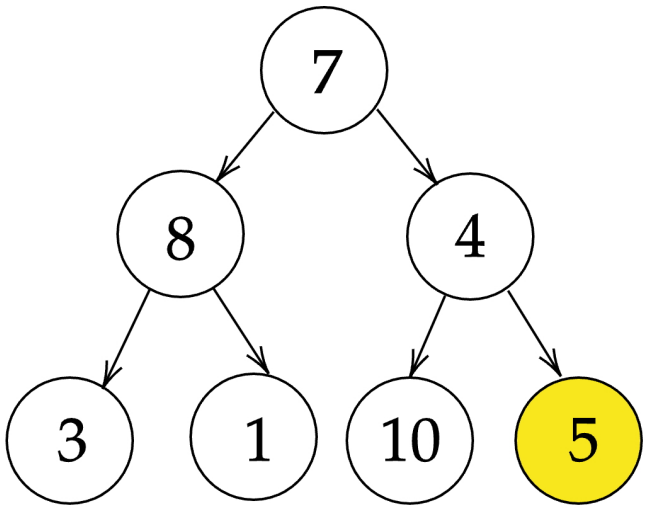
$$\widetilde{A}^\top C_1 A = \alpha_2 \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \; A^\top C_2 A = \alpha_2 I_S, \; \widetilde{S}^\top C_3 \widetilde{S} = \alpha_2 \begin{pmatrix} c_1 & -c_2 \\ -c_1 & c_2 \end{pmatrix}.$$

When $a \in (0, 1], b_1 > 0, b_2 \in (0, a/2)$ *and* $c_1 > 0$, $c_2 \in (0, 1/2)$, *we have* $\widehat{O}_{r2g}(\mathcal{T}; \theta) \to O_{r2g}(\mathcal{T})$ *as* $\alpha_1, \alpha_2 \to +\infty$.

- Ensures the feature interplays to enable autonomous switching.

Assumptions for training dynamics

Training distribution: Let $m \geq 3$ and $S \geq 2^{m+1} - 1 =: N$. The training distribution P_{train} is the **uniform distribution** over **perfect binary trees** $\mathcal{T} = (\mathcal{V}(\mathcal{T}), \mathcal{E}(\mathcal{T}), g(\mathcal{T}))$ of depth m with distinct nodes chosen from $[S]$ and a goal node $g(\mathcal{T})$ being one of the leaf nodes.



Orthonormal embeddings: (i) $\{a_i\}_{i \in [S]}$ are orthonormal vectors in $\mathbb{R}^{d_1}$; (ii) s_f, s_b are orthonormal vectors in $\mathbb{R}^{d_2}$.

Training objective

Autoregressive supervised learning: at each step $k \geq 2$, the input

$$E_{\text{train}}^{(k-1)}(\mathcal{T}) = (E_{\text{train}}^{(k-2)}(\mathcal{T}), o^{(k-1)}(\mathcal{T}))$$

is the concatenation of the input at step $k-1$ and the $(k-1)$-th column of the ground truth output. The output is

$$\widehat{o}_{\text{train}}^{(k)}(\mathcal{T};\theta) = f(E_{\text{train}}^{(k-1)}(\mathcal{T});\theta)_{:,-1}.$$

CoT training loss:

$$\mathcal{L}_{\text{train}}(\theta) = \frac{1}{2}\,\mathbb{E}_{\mathcal{T}\sim P_{\text{train}}} \|O(\mathcal{T}) - \widehat{O}_{\text{train}}(\mathcal{T};\theta)\|_{\text{F}}^2.$$

Gradient descent: we initialize all parameters to be zero, and

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta \mathcal{L}_{\text{train}}(\theta^{(t)}), \quad \forall t \geq 0.$$

Theory of optimization

Theorem (Training dynamics)

We use zero initialization. For learning rate η and ϵ that are sufficiently small, we reach

$$\mathcal{L}_{\text{train}}(\theta^{(t)}) \leq \epsilon$$

as soon as the number of iterations satisfies

- *For backward reasoning*

$$T = \mathcal{O}\left(\frac{mS}{\eta\epsilon}\log\left(\frac{Nm}{\epsilon}\right)\right).$$

- *For forward reasoning*

$$T = \mathcal{O}\left(\frac{S}{\eta}\left(\frac{m}{\epsilon}\right)^{3/2}\log\left(\frac{N}{\epsilon}\right)\right).$$

- Backward reasoning converges at a rate of $\widetilde{\mathcal{O}}(1/\epsilon)$, and forward reasoning converges at a rate of $\widetilde{\mathcal{O}}(1/\epsilon^{3/2})$.

Theory of generalization

Generalization error: on unseen tree $\mathcal{T}$

$$\mathcal{L}_{\text{test}}(\mathcal{T};\theta) = \frac{1}{2}\|O(\mathcal{T}) - \widehat{O}(\mathcal{T};\theta)\|_{\text{F}}^2.$$

Theorem (Generalization)

Given any tree $\mathcal{T}$ at test time with $\widetilde{N}$ distinct nodes chosen from $[S]$ and a path length $\widetilde{m}$, there exists small enough $\epsilon_0 = \epsilon_0(m, \widetilde{N}, \widetilde{m}) > 0$ such that for any $\epsilon \in (0, \epsilon_0]$, when the training less $\mathcal{L}_{\text{train}}(\theta^{(t)}) \le \epsilon$,

- *For backward reasoning,*

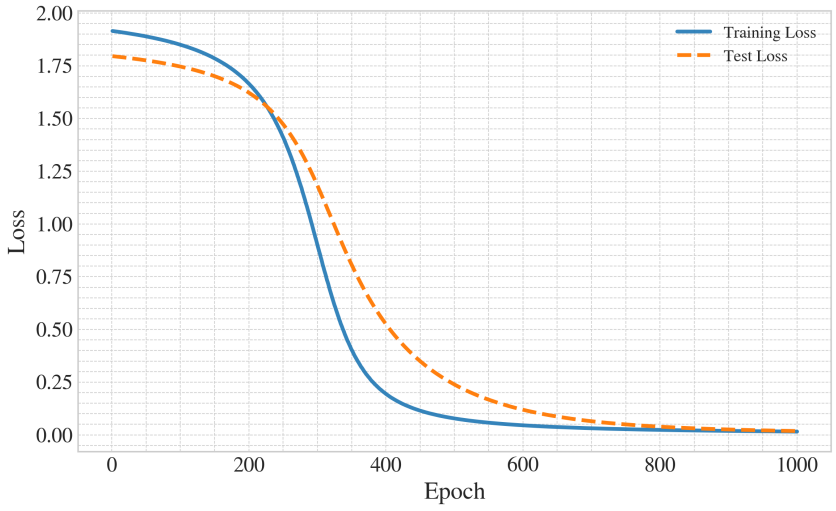
$$\mathcal{L}_{\text{test}}(\mathcal{T};\theta^{(t)}) \le 4\max\left\{\left(\frac{\widetilde{N}+\widetilde{m}-1}{N+m-2}\right)^2, 1\right\}\cdot\frac{\widetilde{m}}{m}\epsilon.$$

- *For forward reasoning,*

$$\mathcal{L}_{\text{test}}(\mathcal{T};\theta) \le 4\max\left\{\left(\frac{\widetilde{N}+2\widetilde{m}-1}{N+2m-1}\right)^2, \left(\frac{\widetilde{N}}{N}\right)^2, 1\right\}\epsilon.$$

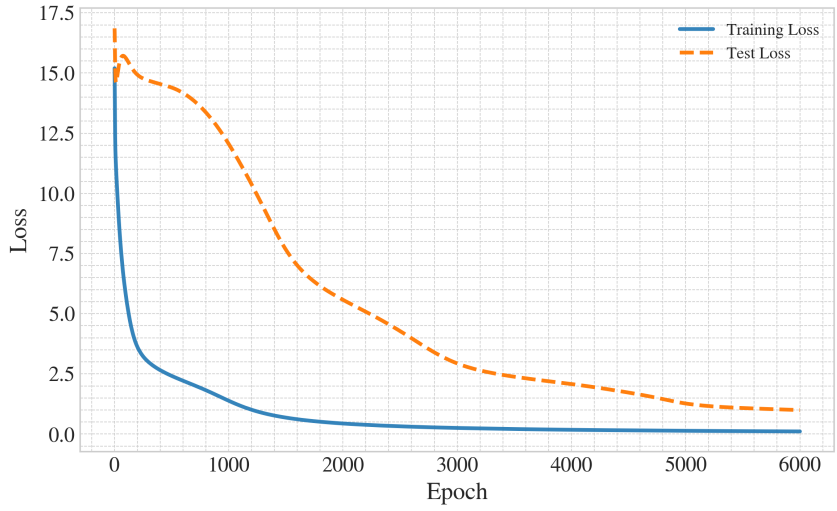
- Better length generalization on unseen trees for small training error.

Numerical experiments



Backward reasoning

$m = 4,\ S = 31$



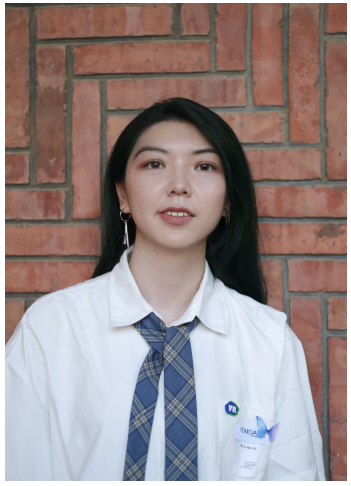
Forward reasoning

$m = 3,\ S = 25$

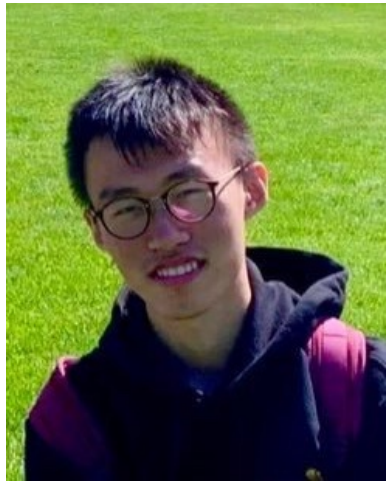
- The training dynamics of attention matrices also follow our analysis.

Transformer generalizes, rather than memorizes.

Transformers learn multi-step compositional reasoning with length generalization



Yu Huang*
Penn



Zixin Wen*
CMU

Aarti Singh
CMU

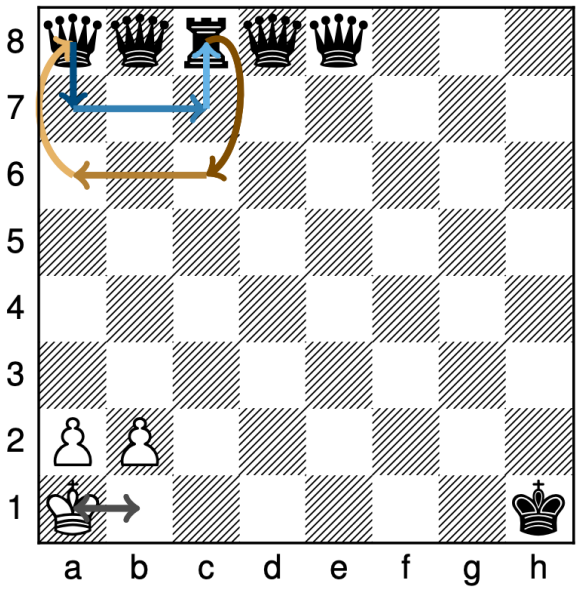
Yuxin Chen
Penn

"Transformers Provably Learn Chain-of-Thought Reasoning with Length Generalization,"
Y. Huang*, Z. Wen*, A. Singh, Y. Chi, Y. Chen, *NeurIPS 2025*

State tracking

Given an initial state and a sequence of *actions*, compute the final state.

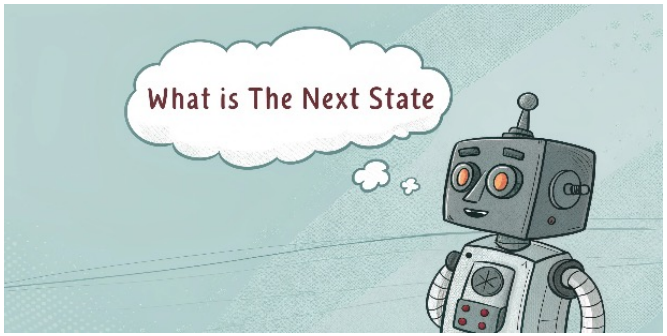
Playing Chess



Images modified from Merrill et al., 2024

Code evaluation

```
x = [0, 0, 1, 0, 0]
x[1], x[3] = x[3], x[1] # Swap 1, 3
```



Entities tracking

Alice, Bob, Carl, Dan, and Emma each have a coin. All are dimes except Carl's. Alice and Carl trade coins.

State tracking task: LEGO

LEGO sentence (Zhang et al., 2022): with answer up to $m < L$

$$\underbrace{\boxed{x_1 = g_1(x_0)}, \ \ldots, \ \boxed{x_L = g_L(x_{L-1})}}_{\text{predicate clauses}} \qquad \underbrace{\boxed{x_0 = y_0}, \ \ldots, \ \boxed{x_m = y_m}}_{\text{answer clauses}}$$

with action $g_i \in \mathcal{G}$, value $y_i \in \mathcal{Y}$, variable $x_i \in \mathcal{X}$.

Goal: correctly calculate the last value $y_L = g_L(g_{L-1}(\cdots g_1(y_0)))$.

$$x_0 \xrightarrow{g_1} x_1 \xrightarrow{g_2} x_2 \xrightarrow{g_3} \cdots \xrightarrow{g_L} x_L, \quad \text{starting with } x_0 = y_0.$$

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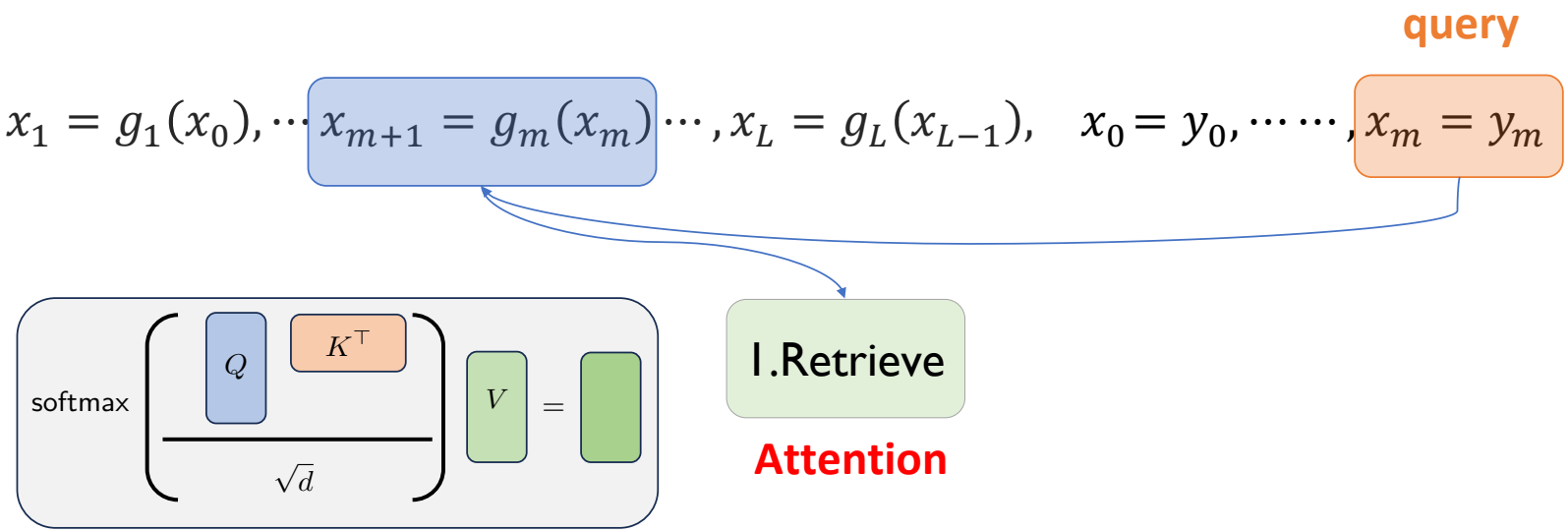
Example: modulo sum

$$x_1 = x_0 \oplus_5 4, \ x_2 = x_1 \oplus_5 2, \quad x_0 = 0, x_1 = ?, x_2 = ?$$

where

$$x_2 = x_1 \underbrace{\oplus_5 2}_{g_2} = (x_0 \underbrace{\oplus_5 4}_{g_1}) \oplus_5 2 = (\underbrace{0}_{y_0} \oplus_5 4) \oplus_5 2 \equiv 1 \pmod 5.$$

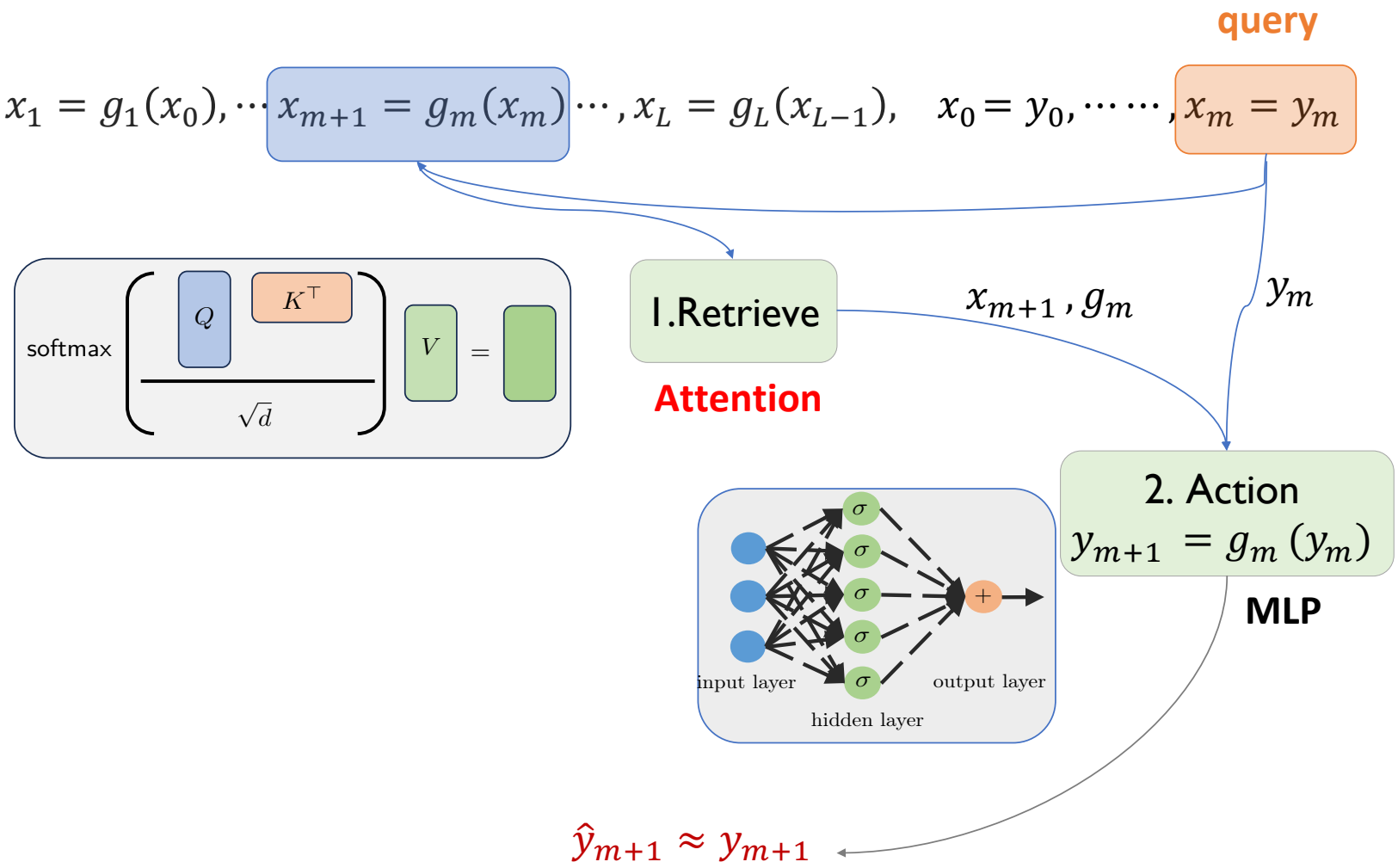
One-layer attention + MLP model



$$\hat{y}_{m+1} \approx y_{m+1}$$

Model mechanism: Attention (retrieval)

One-layer attention + MLP model



Model mechanism: FFN (action) ∘ Attention (retrieval)

Tracking the attention dynamics

$$\mathbf{Z}^{L,m} \triangleq \underbrace{\boxed{x_1 = g_1(x_0)}}_{\mathbf{Z}_{\text{pred},1}}, \ \ldots, \ \boxed{x_L = g_L(x_{L-1})} \quad \underbrace{\boxed{x_0 = y_0}}_{\mathbf{Z}_{\text{ans},0}}, \ \ldots, \ \underbrace{\boxed{x_m = y_m}}_{\text{query } \mathbf{Z}_{\text{ans},m}}$$

Training objective: minimize the cross-entropy loss for *next-answer* prediction via gradient descent

$$\mathcal{L}(F) = \mathbb{E}\Big[\sum_{m=1}^{L} -\log p_F(y_{m+1} \mid \mathbf{Z}^{L,m})\Big]$$

where $d = |\mathcal{X}| + |\mathcal{G}| + |\mathcal{Y}|$ (roughly embedding dimension).

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Uniform attention initialization: $\mathbf{Attn}_{\text{ans},m\to\mathbf{k}}$ is the same for all clauses

- $\mathbf{Attn}_{\text{ans},m\to\mathbf{k}}$: attention score from $\mathbf{Z}_{\text{ans},m} \to \mathbf{Z}_{\mathbf{k}}$;

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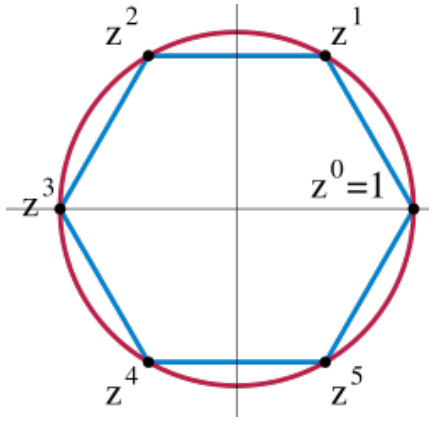
Tracking how attention concentrates on the key clauses and **quantify** the concentration *at loss convergence*

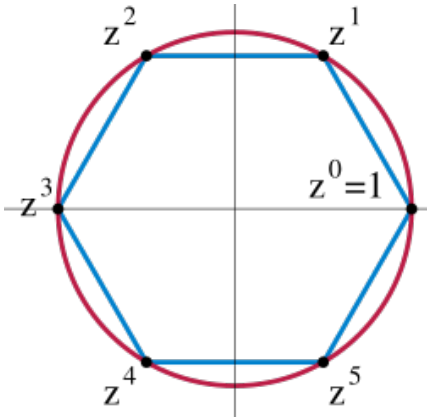
Two group actions

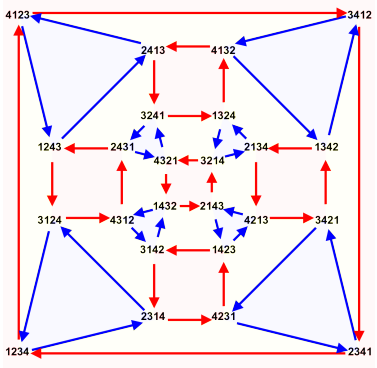


Simple transitive action: transitive and $\forall y, y'$, there is a unique $g \in \mathcal{G}$ s.t., $g(y) = y'$

- Cyclic group (mod n)
- The task is in TC^0

Two group actions





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Symmetry action S_n: all permutations (bijections) of n elements, with composition as the operation

- $\mathcal{G}_{y \to y'} = \{g \in \mathcal{G} : \ g(y) = y'\}$ is large
- S_n with $n \geq 5$ is in NC^1 (harder)

Length generalization for simple transitive group

Theorem (Short-to-poly length generalization)

If $\mathcal{G}$ is simply transitive, then the model trained via GD on the LEGO task of $L \leq 2$ can directly solve tasks for $L \leq \mathrm{poly}(d)$.

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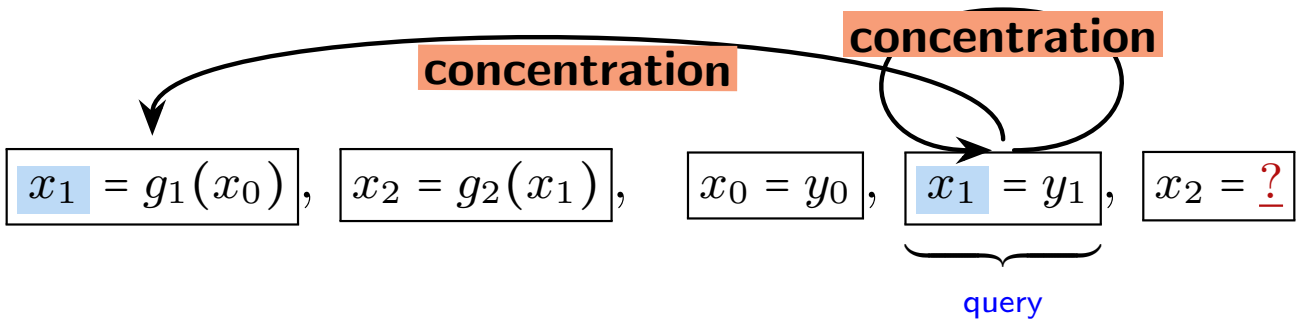
Mechanism: FFN perfectly captures action; while attention learns to concentrate on the **relevant** clauses.



Attention concentration: $\mathbf{Attn}_{\text{ans},\ell\to\text{pred},\ell} + \mathbf{Attn}_{\text{ans},\ell\to\text{ans},\ell} = 1 - \frac{1}{\mathrm{poly}\, d}$

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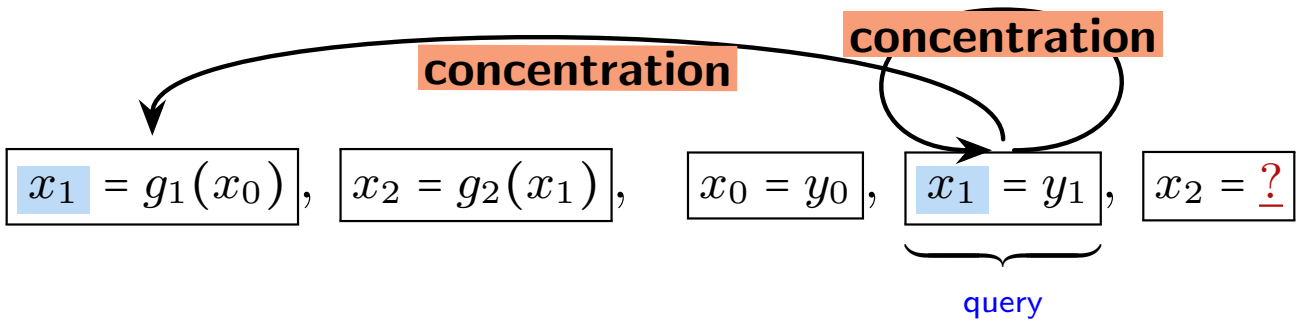

align the shared variable x_ℓ.

Attention concentration: $\mathbf{Attn}_{\mathsf{ans},\ell\to\mathsf{pred},\ell} + \mathbf{Attn}_{\mathsf{ans},\ell\to\mathsf{ans},\ell} = 1 - \frac{1}{\mathrm{poly}\, d}$

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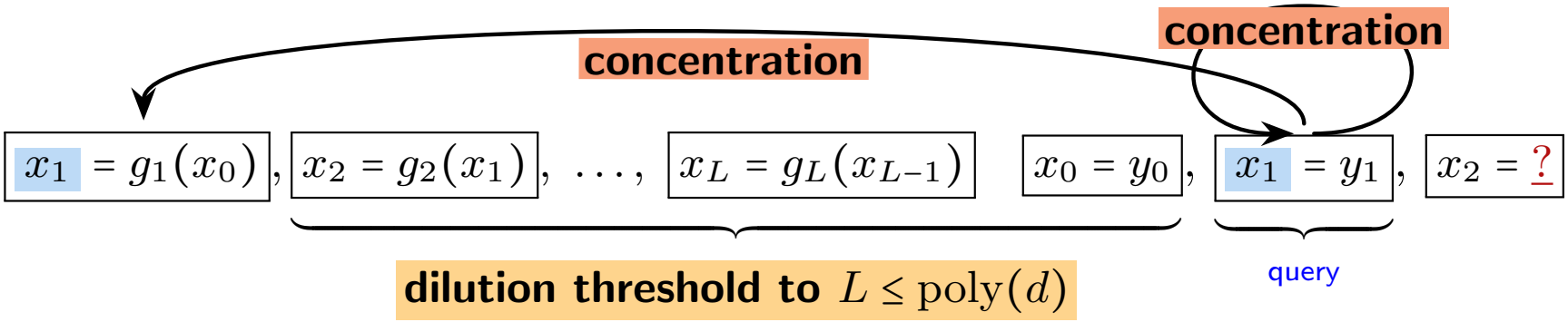

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Length generalization for symmetry group

Hardness from symmetry $|\mathcal{G}_{y \to y'}| \gg 1$, simply transitive $|\mathcal{G}_{y \to y'}| = 1$.

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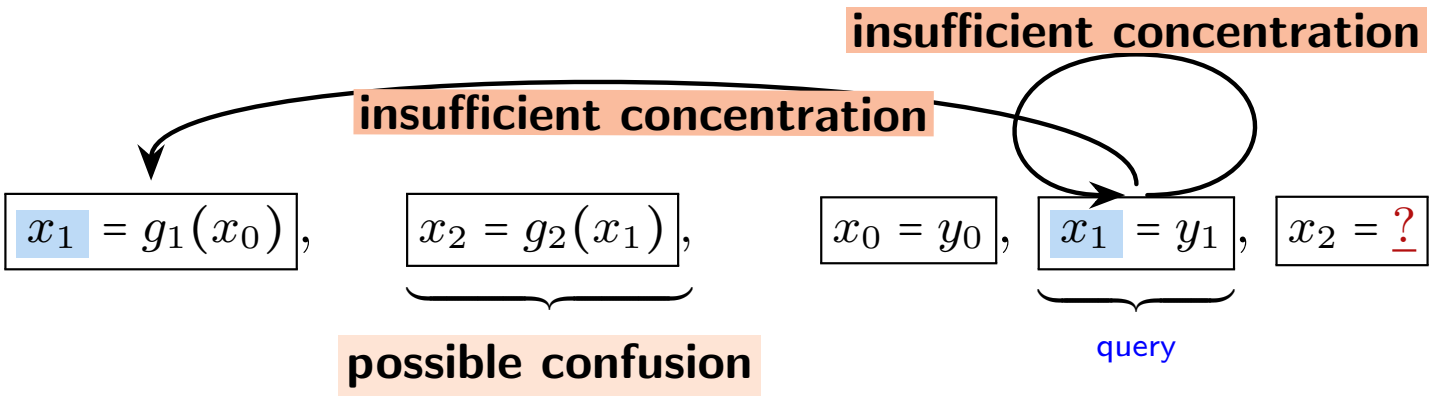
Predicate lookup faces many **near-duplicates**:



$$\mathbf{Attn}_{\text{ans},\ell\to\text{pred},\ell} + \mathbf{Attn}_{\text{ans},\ell\to\text{ans},\ell} = 1 - \text{small constant}$$

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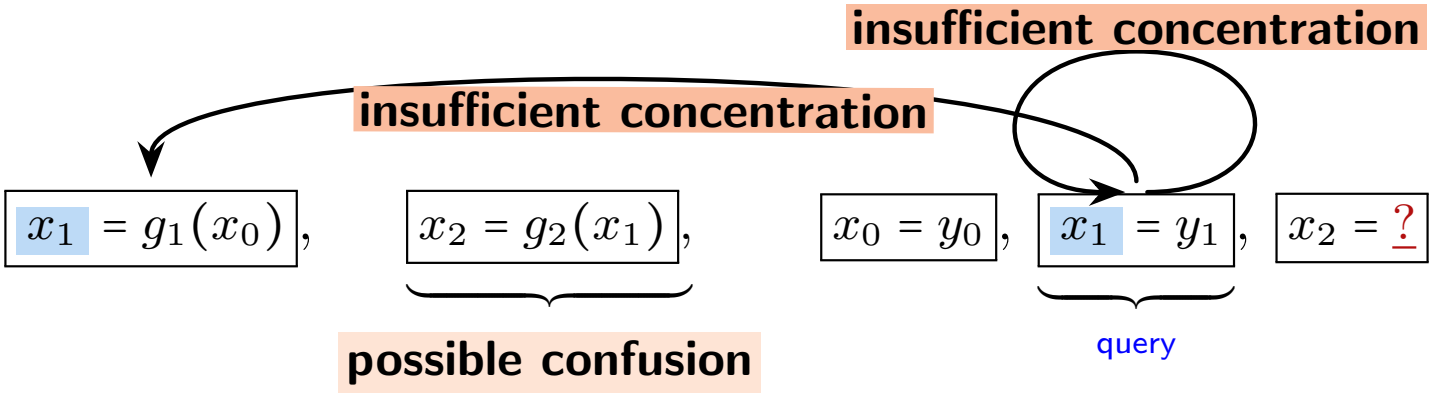
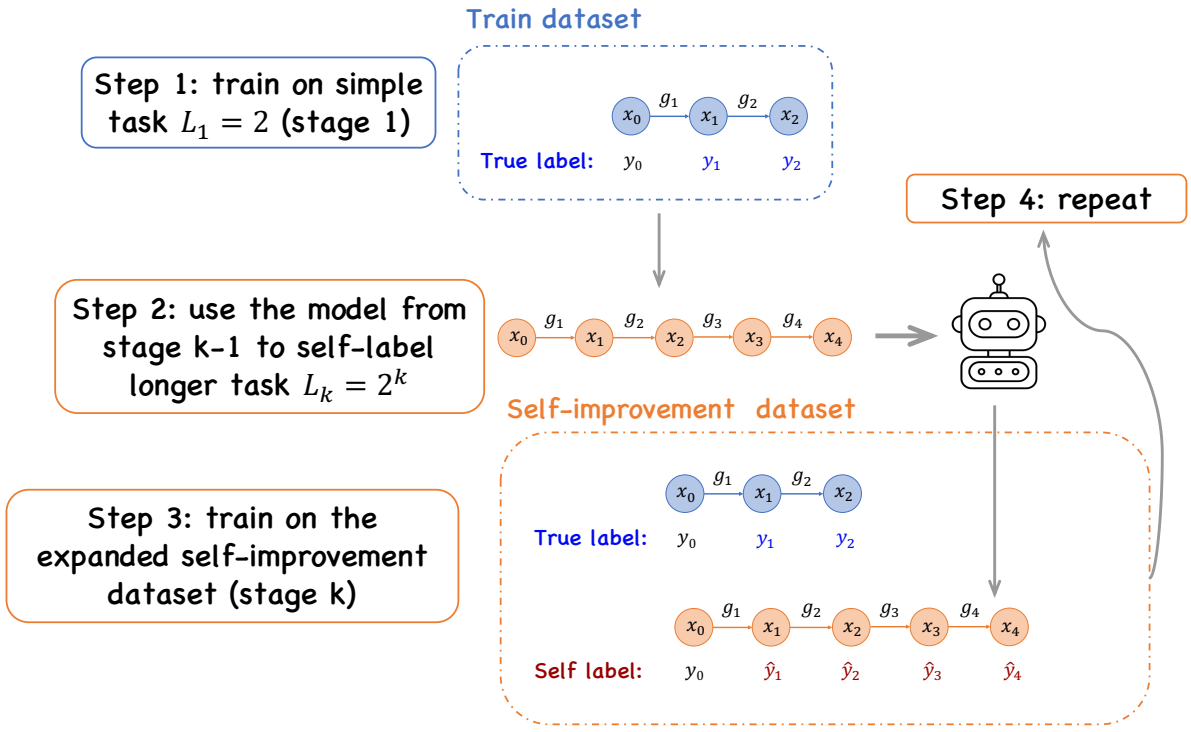


$$\mathbf{Attn}_{\text{ans},\ell \to \text{pred},\ell} + \mathbf{Attn}_{\text{ans},\ell \to \text{ans},\ell} = 1 - \text{small constant}$$

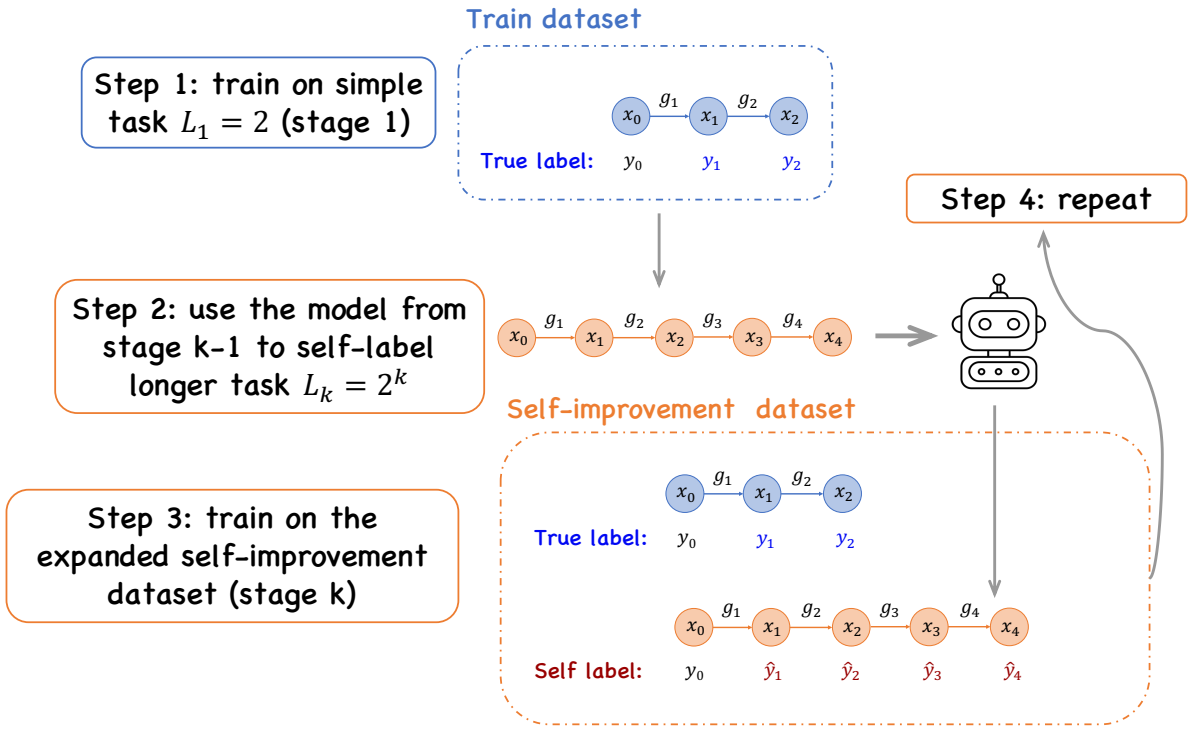
No guarantee beyond **constant**-factor length generalization!

Self-improvement provably helps



Inspired by (Lee et al., 2025)

Self-improvement provably helps



Inspired by (Lee et al., 2025)

Theorem (Self-improvement for length generalization)

If $\mathcal{G}$ is symmetry (S_n), then the model trained via self-training on $L = 1, 2, \cdots, 2^{k-1}$ can generalize to solve task of $L = 2^k$. After $\Theta(\log d)$ stages, it can solve tasks of $L \leq d$.

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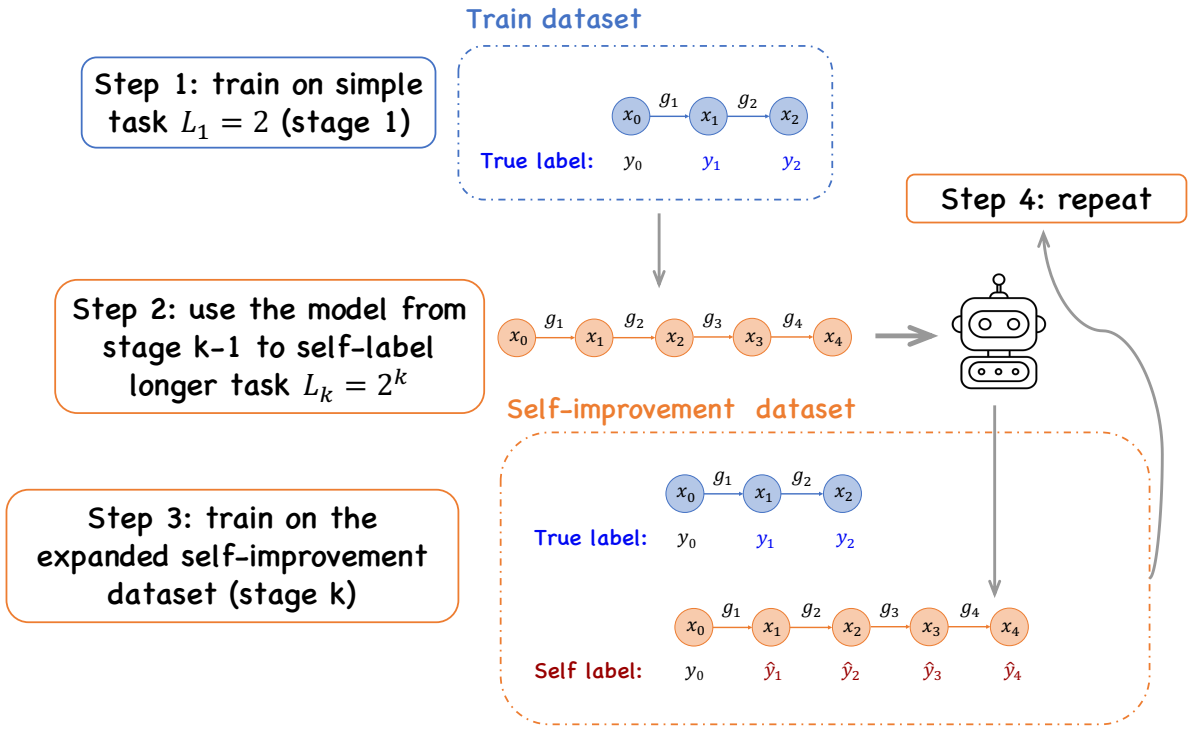

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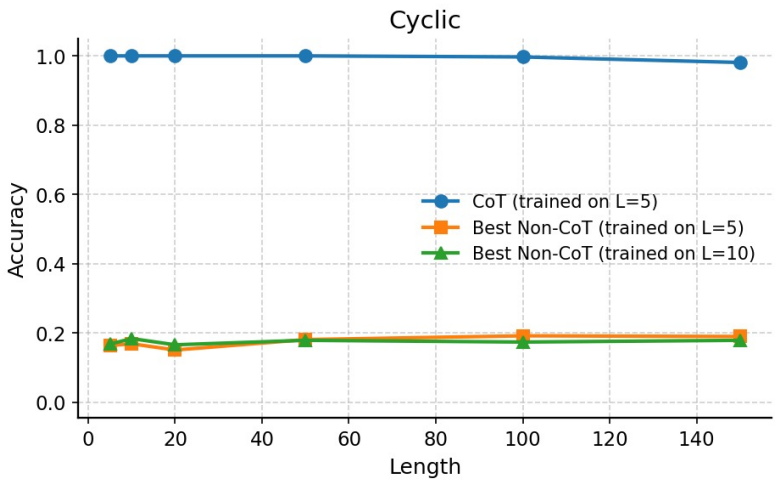
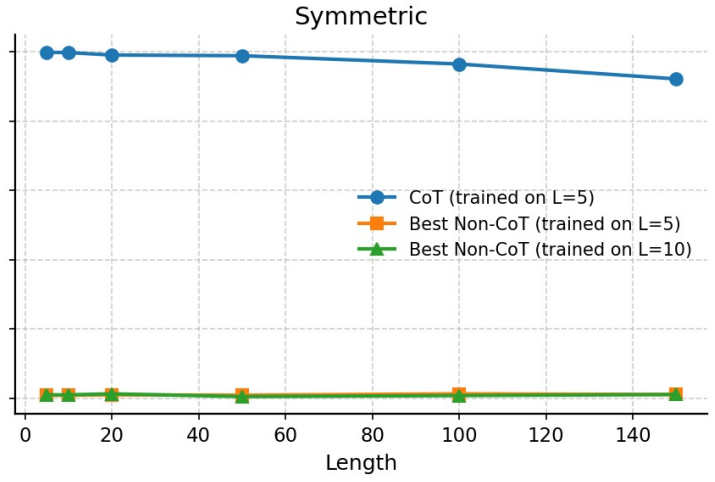
Transformer with CoTs **learns** to solve problems in NC^1!

Numerical experiments



Cyclic group (C_6)

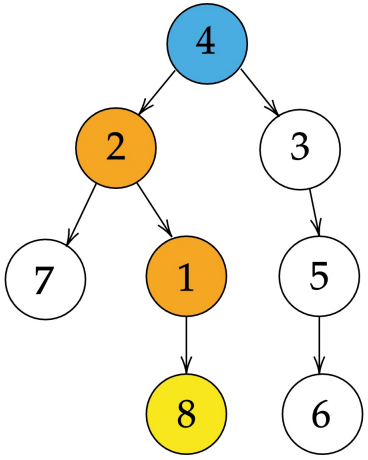


Symmetric group (S_5)

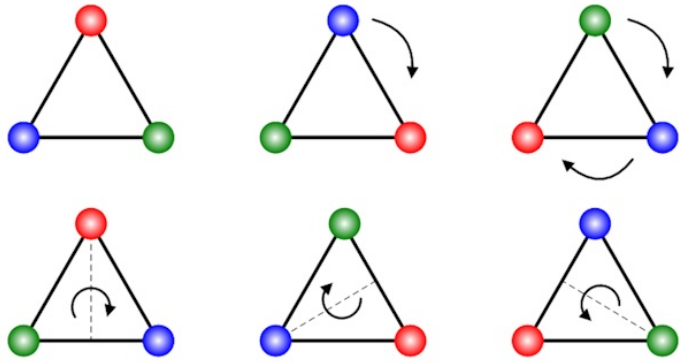
Transformer with CoTs **generalizes** much better than non-CoT training

Summary

symbolic reasoning



reasoning with group action



Take-away: one-layer transformers with CoT training can reason with length generalization (without positional encoding)

Future work:

- Other reasoning tasks?
- Understanding RL for CoT reasoning.

Thanks!

- Multi-head Transformers Provably Learn Symbolic Multi-step Reasoning via Gradient Descent, NeurIPS 2025, arXiv 2508.08222.
- Transformers Learn Chain-of-Thought Reasoning with Length Generalization, NeurIPS 2025, to be arXived soon.

https://yuejiechi.github.io